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Project Title:	Mechanisms for Plant Adaptation to Space Environment		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Task Description:

Successful plant growth in closed-loop life support conditions is a difficult challenge for the realization of long-term habitation of spacecraft and other extraterrestrial environments. In such environments, plants can undergo stress induced by a number of factors including changes in gravity, radiations, vibration, limited exchange of gases, and suboptimal growth conditions (temperature, light, nutrients). These sources of stress are often associated with reprogramming of gene expression and can cause limited plant growth, development, and yield. To facilitate plant life in space, it is crucial to acquire a better understanding of the genetic changes that enable plant cells to respond to spaceflight stress. To do so, one goal of this proposal is to define the underlying mechanisms of plant adaptation to the spaceflight environment at a transcriptional level. Through research supported by NASA, we have successfully identified a protein named AtIRE1 as a master regulator of transcription in conditions of stress responses to abiotic, biotic stress, and to gravity changes in plants. However, the downstream signaling pathways controlled by AtIRE1 are not well defined, especially in conditions of altered gravity. A better understanding of the signaling pathways controlled by AtIRE1 is important to understand how plants can grow in conditions of stress. We propose to use in-flight and ground resources along with genomics and transcriptomics analyses in the model plant Arabidopsis thaliana to understand the regulatory role of AtIRE1 in adaptation to spaceflight stress. Development of our research will continue to contribute to the understanding of basic signaling pathways that are in place to ensure stress survival in hostile environments, thus making possible the design and growth of plants that are resistant to space stress. To contribute further to the successful realization of habitation in space, we also aim to develop plants that can function as bioindicators of stress during in-flight situations. To do so, we will engineer plants with an AtIre1 substrate that is activated specifically in conditions of stress and that we will adapt to function as a visual stress reporter. The availability of real-time stress bioindicators will provide scientists and astronauts with direct read-outs of stress in the space environment. The results gathered in our research will contribute to NASA's strategic plans for the realization of long-term habitation of space and planetary surfaces. Because of the conservation of stress sensing and response mechanisms across multicellular organisms, we expect that our results will also have important implications for the general knowledge of stress and in the design of solutions for space stress management in multicellular organisms, including humans.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The molecular study of human health conditions that develop in response to long term habitation in spaceflight conditions is inherently limited by the necessity to insure safety of study participants. As such, the cellular and molecular pathology of these disease states and potential remediations are unknown. Comparative biology, or the study of similar situations in different types of organisms, presents a unique opportunity to explore these human conditions with minimal associated risks. The study of conserved cellular stress responsive pathways between humans and plants provides this benefit and many others. In particular, plants are the primary production source for metabolic energy and oxygen in our terrestrial environment, and proper application in extraterrestrial and spaceflight conditions could potentially reduce the cost of recreating human habitable environments in extremely harsh conditions.
Plants, like animals, suffer from stress linked to a variety of challenges. For example, certain adverse environmental conditions, such as changes in temperature, reduced nutrients, and disease, as well as physiological situations, such as growth, are known to generate an imbalance between demand and capacity of protein synthesis in the endoplasmic reticulum (ER), a condition that is also known as ER stress (1, 2). The ER, a conserved structure in human and plant cells, initiates the production of all secretory proteins. Secretory proteins constitute one third of the cellular proteome and allow cells to communicate with the external environment. The ER can sense stress due to accumulation of unfolded proteins in its lumen, and restores homeostasis by invoking a largely conserved cytoprotective signaling pathway, called the unfolded protein response (UPR). Compromised UPR leads to serious conditions and even death in animals and plants (3-5). In spaceflight conditions a number of factors including microgravity, radiations, vibration, limited exchange of gases, and suboptimal growth conditions (temperature, light, nutrients) reduces overall plant growth, impairs development, and reduces yield.

In animals and plants, several sensors and transducers initiate the UPR; however, IRE1, an ER-associated protein kinase and ribonuclease, functions as a major ER stress sensor and plays a key role in the UPR signaling by controlling UPR gene expression (4, 6, 7). Activities in our lab and others have demonstrated that plant IRE1 is needed to respond to a diversified portfolio of stimuli linked to abiotic and biotic stress and cell growth and differentiation, as well as to modulate responsive outputs (4, 5).

In all eukaryotes, IRE1 activates the UPR by an unconventional splicing of a bZIP transcription factor, namely Hac1 in yeast (8-10) and Xbp1 in metazoans (11, 12). Recent efforts in our lab and others have shown that in the model species Arabidopsis thaliana, the functional equivalent of Hac1 and Xbp1 is bZIP60 (5, 13, 14). The bZIP60 mRNA splicing is followed by the synthesis of the mature form of bZIP60, which can modulate transcription of UPR genes in the nucleus. Thus, synthesis of the transcriptionally active bZIP60 occurs only in conditions of stress. In our research we will adopt the stress-dependent activation of bZIP60 to generate plants that can function as real-time stress bioreporters that can be easily used to report on the onset of stress in plants.

We believe that we will learn the mechanisms that enable plants to respond to ER stress. This is highly significant for the understanding of basic defense mechanisms in plants, and for engineering plants that can cope with stress on Earth.

We also expect that our research will have an important impact in biology with implications well beyond plants. Because the signaling pathways between plant and mammalian UPR are largely conserved, it is likely that understanding the UPR in a tractable multicellular model system, such as Arabidopsis, will allow building approaches with a strong potential to contribute to the solution of the problems in humans confronting cellular stress.

We also expect that we will be able to develop transplantable bioindicators of stress without the need of sophisticated instrumentation. These reporters will potentially facilitate the optimization of plant cultivation in hostile environments, such as marginal lands. In addition, because of the conservation of UPR mechanisms in plants and humans, real-time stress bioindicators will likely reflect the perception of stress in other organisms besides plants and thus they may be useful to develop practices to improve quality of life for humans.

In conclusion, not only will our research improve our nation's scientific competitiveness, it will also benefit education and the quality of life on Earth. Life on Earth depends on plants. With sharply growing environmental concerns (global warming, pollution, necessity to expand agriculture to marginal lands, etc.), understanding how plants can resist stress in hostile environments and adopting bioindicators help research to expand agriculture practices and thus ensure continued survival of the ever-growing human population.

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Task Progress:

Our project focuses on understanding plants' responses to space stress to improve their performance in spaceflight. We focus on proteotoxic stress responses – collectively called the unfolded protein response (UPR). In our previous BRIC-18-2 (Biological Research in Canisters) experiment, we established differential gene regulation in wild type but also in an Arabidopsis UPR mutant. These findings pave the ground to our current experiment and to new areas of research on plant management of the different stresses that are found in space. We are expanding upon our original findings through exploration of a larger set of Arabidopsis UPR mutants that have been flown to the International Space Station (ISS) in a BRIC-22 experiment. These mutants are missing genes that are critical in regulating the response to stresses observed previously, but have different molecular functions. We successfully completed both of the preflight tests (science verification test (SVT) and experiment verification test (EVT)) to validate the exact experimental protocols necessary to prepare and analyze the samples returned from BRIC-22.
On August 14, 2017 our experiment was launched on board the SpaceX-12 ISS resupply mission and was successfully executed by ISS astronauts. The samples were preserved on orbit and underwent atmospheric reentry on board the same Dragon capsule in September of 2017. These were then returned to the Brandizzi lab still preserved at -80 degrees C. The samples were processed yielding experimental material that was determined to be in excellent condition. The RNA extracted from these plants has recently completed the high throughput sequencing process. A large data set comprising a total of 1.45 billion reads (yield of 72.77 Giga base pairs) and an average of approximately 32 million reads per sample was acquired. More than 95% of all raw reads were successfully mapped to the Arabidopsis genome for each sample. We identified differentially expressed genes using a standard bioinformatics pipeline. For each genotype, differentially expressed genes in flight samples relative to ground samples were determined using a strict criteria (adjusted P value <0.01; 0.5 > Fold Change >2). In total, more than 4,000 differentially expressed genes were found to be up or down regulated in at least one of the tested genotypes. In wild type plants more than 3000 genes (>2 fold change) were upregulated under spaceflight conditions. Additionally, we found that around 300 genes were downregulated (>0.5 fold) in wild type plants in flight samples compared to ground. To help dissect the biological implications of this large data set, we have thus far established a baseline understanding of what known biological processes were represented by the transcripts altered in the wild type background in response to space flight. To do so we utilized PANTHER Gene Ontology (GO) term enrichment analysis (PANTHER, Overrepresentation Test, Fisher's Exact test type with False Discovery Rate calculation; corrected p-value <0.05). In wild type, over 100 GO terms were either significantly enriched or surprisingly underrepresented in the 3000 upregulated differentially expressed genes. Furthermore ~50 GO terms were significantly enriched in the downregulated differentially expressed genes. We are currently extending this analysis to better understand how UPR regulators contribute these observations by comparing patterns of gene regulation in UPR mutant lines to our findings in wild type.

Additionally, further research on the genes missing from the Arabidopsis mutants to be tested in BRIC-22 is also being completed in ground based studies. Through a better understanding of gene function at a cellular and organism level in terrestrial growth conditions, we will be better positioned to explore what role these genes have in spaceflight conditions. This information will be adapted to help produce Arabidopsis lines resistant to these stresses.

Bibliography Type: Description: (Last Updated: 12/08/2025)

Articles in Peer-reviewed Journals	Angelos E, Brandizzi F. "NADPH oxidase activity is required for ER stress survival in plants." Plant J. 2018 Dec;96(6):1106-20. Epub 2018 Oct 23. https://doi.org/10.1111/tpj.14091 ; PubMed PMID: 30218537; PubMed Central PMCID: PMC6289879 , Dec-2018